W02 DRINKING WATER QUALITY | P

Intent : Provide access to drinking water that complies with health-based limits on chemical composition.

Summary : This WELL feature requires projects to provide drinking water that meets thresholds on chemicals as published by research and regulatory organizations.

Issue : The chemical composition of drinking water, and therefore, its quality, changes from city to city and even within buildings due to the highly variable conditions of its sourcing, treatment and distribution within cities and inside buildings.[1] For example, natural deposits have caused arsenic to leach into some groundwaters to reach levels above drinking water health guidelines.[2] Water streams can also pick up contaminants from agricultural runoffs and direct industrial discharges,[3] whereas drinking water may encounter many opportunities to pick up contaminants in its travel from a treatment plant to the point of use, including corrosion byproducts such as lead and copper.[1] Disinfectants used to prevent microbial growth and render water potable, such as chlorine, may react with natural organic matter and yield unwanted disinfectant byproducts (DBPs) such as trihalomethanes (THMs) and haloacetic acids (HAAs), to which chronic exposure needs to be minimized.[4]

Solutions : Drinking water is treated and distributed to meet applicable legal requirements and regulations that may differ by country, [5] and many building-scale interventions can improve water quality depending on the contaminants that need to be removed. Typical technologies able to capture contaminants include activated carbon filters, ion exchange resins and reverse osmosis (RO) systems. Evaluating chemical parameters such as pH and free chlorine may inform of the potential for the uptake of corrosion byproducts and/or bacterial growth in drinking water.[1]

PART 1 MEET CHEMICAL THRESHOLDS

For All Spaces:

The following requirements are met:

a. The project provides at least one drinking water dispenser, plus one drinking water dispenser per dwelling unit (if applicable).

b. Drinking water dispensers provide water that meets the following parameters:[1]

1. Arsenic ≤ 0.01 mg/L.
2. Cadmium ≤ 0.003 mg/L.
3. Chromium (total) ≤ 0.05 mg/L.
4. Copper ≤ 2 mg/L.
5. Fluoride ≤ 1.5 mg/L.
6. Lead ≤ 0.01 mg/L.
7. Mercury (total) ≤ 0.006 mg/L.
8. Nickel ≤ 0.07 mg/L.
9. Nitrate ≤ 50 mg/L as Nitrate (11 mg/L as Nitrogen).
10. Nitrite ≤ 3 mg/L as Nitrite (0.9 mg/L as Nitrogen).
11. Total chlorine ≤ 5 mg/L.

c. Drinking water dispensers provide water that meets the following parameters:

1. Residual (free) chlorine does not exceed 4 mg/L.[3]
2. The concentration of total trihalomethanes (TTHM, sum of dibromochloromethane, bromodichloromethane, chloroform and bromoform) is 0.08 mg/L or less.[3]
3. The concentration of haloacetic acids (HAA5, sum of chloroacetic, dichloroacetic, trichloroacetic, bromoacetic and dibromoacetic acids) is 0.06 mg/L or less.[3]

Note : Multifamily residential projects may achieve WELL Certification at the Bronze or Silver level without testing in dwelling units, but cannot achieve Gold or Platinum without testing in dwelling units. See Sampling Rates for Multifamily Residential in the WELL Performance Verification Guidebook for further details. Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

PART 2 MEET THRESHOLDS FOR ORGANICS AND PESTICIDES

For All Spaces:

Option 1: Drinking water quality report

The following requirements are met:

a. A municipal water quality report issued not more than one year before enrollment or the start of subscription reports on at least two of the pesticides below. All reported pesticides within the municipal water quality report comply with the following thresholds or are found to be compliant through Option 2, On-Site Testing:[1]

1. Aldrin and Dieldrin (combined): 0.00003 mg/L or less.
2. Atrazine: 0.1 mg/L or less.
3. Carbofuran: 0.007 mg/L or less.
4. Chlordane: 0.0002 mg/L or less.
5. 2,4-Dichlorophenoxyacetic acid (2,4-D): 0.03 mg/L or less.
6. Dichlorodiphenyltrichloroethane (DDT) and metabolites: 0.001 mg/L or less.

7. Lindane: 0.002 mg/L or less.
8. Pentachlorophenol (PCP): 0.009 mg/L or less.

b. A municipal water quality report issued not more than one year before enrollment or the start of subscription reports on at least three of the organic contaminants below. All reported organic contaminants within the municipal water quality report comply with the following thresholds or are found compliant through Option 2, On-Site Testing:[1]

1. Benzene: 0.01 mg/L.
2. Benzo[a]pyrene: 0.0007 mg/L.
3. Carbon tetrachloride: 0.004 mg/L.
4. 1,2-Dichloroethane: 0.03 mg/L.
5. Tetrachloroethene (Tetrachloroethylene): 0.04 mg/L.
6. Toluene: 0.7 mg/L.
7. Trichloroethene: 0.02 mg/L.
8. 2,4,6-Trichlorophenol: 0.2 mg/L.
9. Vinyl Chloride: 0.0003 mg/L.
10. Xylenes (o-, m- and p-): 0.5 mg/L.

OR

Option 2: On-site testing

The following requirements are met:

a. Testing must be conducted at drinking water dispensers for at least two of the pesticides from the list below. All contaminants in the list below that are reported by the lab must comply with the following thresholds:

1. Aldrin and Dieldrin (combined): 0.00003 mg/L or less.
2. Atrazine: 0.1 mg/L or less.
3. Carbofuran: 0.007 mg/L or less.
4. Chlordane: 0.0002 mg/L or less.
5. 2,4-Dichlorophenoxyacetic acid (2,4-D): 0.03 mg/L or less.
6. Dichlorodiphenyltrichloroethane (DDT) and metabolites: 0.001 mg/L or less.
7. Lindane: 0.002 mg/L or less.
8. Pentachlorophenol (PCP): 0.009 mg/L or less.

b. Testing must be conducted at drinking water dispensers for at least three of the organic contaminants from the list below. All contaminants in the list below that are reported by the lab must comply with the following thresholds:

1. Benzene: 0.01 mg/L.
2. Benzo[a]pyrene: 0.0007 mg/L.
3. Carbon tetrachloride: 0.004 mg/L.
4. 1,2-Dichloroethane: 0.03 mg/L.
5. Tetrachloroethene (Tetrachloroethylene): 0.04 mg/L.
6. Toluene: 0.7 mg/L.
7. Trichloroethene (Trichloroethylene): 0.02 mg/L.
8. 2,4,6-Trichlorophenol: 0.2 mg/L.
9. Vinyl Chloride: 0.0003 mg/L.
10. Xylenes (o-, m- and p-): 0.5 mg/L.

Note : Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

REFERENCES

1. World Health Organization. Guidelines for drinking-water quality. 4th ed. Geneva, Switzerland: WHO Press; 2017.
2. World Health Organization. Arsenic. http://www.who.int/news-room/fact-sheets/detail/arsenic. Published 2018. Accessed 2020.
3. U.S. Environmental Protection Agency. National primary drinking water regulations. https://www.epa.gov/ground-water-and-drinking-water/national-primary-drinking-water-regulations. Published 2009. Accessed January 20, 2020.

4. World Health Organization. Trihalomethanes in Drinking-water. Geneva, Switzerland2005.
5. World Health Organization. Water Safety in Buildings. Geneva, 2011.